Evaluating the Quantum Energy Advantage: A Breakeven Analysis for Max-Cut Optimization

Gustavo Kerdole Gontijo[1]

[1]Télécom Paris, Institut Polytechnique de Paris, 91120 Palaiseau, France

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Abstract

This project investigated the "Quantum Energy Advantage" by establishing an analytical framework to identify the exact problem complexity threshold where quantum computing becomes more energy-efficient than classical counterparts for a specific optimization task. We bypassed physical hardware testing and instead mathematically modeled total Energy-to-Solution across three paradigms as problem size scales: Classical method and its GPU thermal design power over algorithmic runtime; Noisy Intermediate Scale Quantum (NISQ) systems, factoring in the exponential sampling overhead of quantum error mitigation; and Fault-Tolerant Quantum Computing (FTQC), accounting for fixed cryogenic infrastructure and the heavy classical decoding power required for error correction. By plotting these distinct scaling equations on a unified logarithmic scale, the project pinpointed the precise crossover points where quantum architectures overcome their high initial infrastructure costs to surpass classical power efficiency, providing a clear architectural roadmap for sustainable computation.

1 Introduction

In recent years, the exponential growth of classical artificial intelligence and heavy optimization workflows has triggered a critical sustainability crisis in global data centers[11]. As optimization tasks — such as deep neural network training, portfolio risk management, and logistical network mapping — scale in complexity, their reliance on classical architectures demands massive clusters of high-power graphics processing units (GPUs). The energy consumption of these systems scales aggressively with problem size, bound by the thermal design power of the underlying silicon and the accompanying infrastructure cooling overhead. This relentless trajectory has shifted the primary constraint of high-performance computing from pure execution time to a definitive "power wall," where the financial and environmental costs of energy consumption threaten to outpace computational utility[12].

Concurrently, quantum computing (QC) has emerged as a disruptive alternative, historically celebrated for its theoretical potential to achieve exponential runtime speedups over classical algorithms. However, as the quantum hardware ecosystem matures toward practical deployments, the definition of "Quantum Advantage" is undergoing a critical transformation. The focus is expanding beyond traditional temporal acceleration to encompass a Quantum Energy Advantage[1]. Under this paradigm, the primary metric of interest becomes Energy-to-Solution (ETS)[14] — the total quantity of Joules consumed to successfully resolve a specific computational problem. While individual quantum logic gates consume negligible energy ($\approx 0.001\,\mu W$, see Appendix B) compared to classical transistors ($\approx 4\,\mu W$), quantum architectures require a massive, constant energy overhead to maintain their operational environments, primarily driven by cryogenic cooling and complex classical control electronics (so that in practice, the cryogenic electrical power consumption needed to run the quantum gate is $\approx 1000\,\mu W$.) [7]

This project addresses the vital intersection of these computational trajectories by establishing an analytical Breakeven Analysis framework. Given the short-term constraints of this research project, we bypass physical hardware benchmarking to build a strict mathematical model comparing the scaling behaviors of three distinct computing paradigms: Classical GPU clusters, Noisy Intermediate-Scale Quantum (NISQ) systems, and Fault-Tolerant Quantum Computers (FTQC). To isolate the core algorithmic scaling properties, the analysis focuses strictly on a synthetic, data-independent optimization problem (Max-Cut), thereby eliminating the confounding variable of quantum data-loading bottlenecks. Similar analysis were made for different problems (e.g. breaking RSA encryption [7] and language model fine-tuning [14]), but no work was found focusing on the Max-Cut problem and the energy cost for solving it, justifying our choice. By mathematically mapping the total energy expenditure of each paradigm as a function of problem complexity (N), this study aims to pinpoint the precise crossover thresholds where the long-term architectural efficiency of quantum computing successfully breaks even with, and ultimately surpasses, the scaling limits of classical infrastructure.

2 Quantum Computation and Architectural Paradigms

To establish a rigorous baseline for the energy break-even analysis, it is necessary to outline the core mathematical and physical principles of quantum computing that directly govern algorithmic execution time and hardware power scaling. For a introduction on quantum theory and concepts like quantum state (e.g. a qubit), see Appendix A. This section introduces circuit scaling metrics, and the two primary architectural eras evaluated in this framework.

2.1 Quantum Circuits, Logic Gates, and Circuit Depth

Quantum algorithms are executed by applying a sequence of discrete unitary transformations (quantum gates) to an initial state, followed by measurement (see Appendix A for more details). In superconducting hardware architectures, which is the one we analyze in this work, these operations are physically realized via precise microwave pulses.[13]

The algorithmic complexity — and consequently, the total Energy-to-Solution (ETS), as we will show — is heavily dictated by two structural metrics of the quantum circuit:

- **Gate Count (G):** The total number of quantum operations executed. Two-qubit entangling gates (e.g., $CNOT$ or CZ) are significantly more error-prone and take longer to execute than single-qubit rotations, serving as the primary cost driver.
- **Circuit Depth (D):** The longest path of sequential gate operations from initialization to measurement, assuming ideal parallelization of independent gates, i.e. compression of the circuits. Circuit depth serves as a direct proxy for the physical execution runtime t_{quantum} of a single circuit run, since it's considered all parallel gates run in the same logical time step.

2.2 The Noisy Intermediate-Scale Quantum (NISQ) Paradigm

Contemporary quantum hardware operates firmly within the Noisy Intermediate-Scale Quantum (NISQ) regime, a term introduced by Preskill in 2018 [16], characterized by processors containing tens to hundreds of physical qubits that are not yet large or stable enough to support fault-tolerant error correction. In this era, qubits are physical, imperfect entities highly susceptible to environmental decoherence and operational noise.

In superconducting architectures[13] - such as transmons - the logical states $|0\rangle$ and $|1\rangle$ are mapped to the lowest energy levels of an anharmonic Josephson junction oscillator. Because the energy spacing between these states is minuscule (typically in the microwave regime, $\sim$ 4–6 GHz), the quantum state is extremely fragile. Beyond individual qubit decoherence, multi-qubit systems are severely limited by operational cross-talk. Cross-talk occurs when control signals (microwave pulses) intended for a target qubit parasitically couple to neighboring qubits via stray capacitive or inductive pathways, causing unintended phase shifts and state transitions.

This susceptibility to noise culminates in highly imperfect gate operations. While single-qubit rotations are relatively highly coherent, two-qubit entangling gates (such as the Controlled-Z gates) require driving interaction dynamics over longer physical durations, leaving them highly vulnerable. Modern state-of-the-art superconducting processors exhibit typical physical two-qubit gate error rates on the order of $\epsilon_{\text{phys}} \sim 10^{-3}$[7]. In the absence of error-correcting codes, these errors compound multiplicatively. Consequently, any quantum circuit whose physical gate count G approaches or exceeds $1/\epsilon_{\text{phys}}$ will experience catastrophic noise accumulation, rendering the final measurement distribution indistinguishable from uniform random noise.

To extract meaningful solutions for algorithms like the Quantum Approximate Optimization Algorithm (QAOA), *Quantum Error Mitigation (QEM)* techniques — such as Zero-Noise Extrapolation or Probabilistic Error Cancellation [3] — are required. While QEM avoids physical qubit overhead, it introduces a severe temporal penalty. To statistically reconstruct an error-free expectation value, the number of required measurement samples (shots, M_{shots}) must scale exponentially with the total error profile of the circuit [19]:

$$M_{\text{shots, mitigated}} \propto M_{\text{baseline}} \cdot \gamma^{D \cdot \epsilon_{\text{phys}}} \tag{1}$$

where $\gamma > 1$ is an architecture-dependent mitigation scaling factor. This exponential sampling overhead directly translates into an extended execution runtime, causing the NISQ energy profile to scale aggressively with problem size. So we are interested in seeing how this compares with the other two cases, the classical and FTQC.

2.3 Fault-Tolerant Quantum Computing (FTQC)

To surpass the scaling limitations imposed by the NISQ error wall, large-scale architectures must transition to Fault-Tolerant Quantum Computing (FTQC), first proposed by Peter Shor in 1997 [18]. FTQC bypasses the exponential sampling tax of error mitigation by actively protecting quantum information through error-correcting topological codes, most notably the Surface Code[9], which is the one we consider in our analysis.

In an FTQC architecture, multiple noisy physical qubits are entangled to construct a single, highly stable *logical qubit*. The protection level is determined by the code distance d. The total number of physical qubits required to sustain N logical qubits scales quadratically with d:

$$N_{\text{phys}} = 2d^2 \cdot N \tag{2}$$

While surface codes suppress physical errors exponentially with respect to d, they introduce a significant hardware and energy bottleneck. Error checking requires continuous, real-time classical processing of error syndromes at kilohertz frequencies, demanding substantial classical decoding power (P_{decoder}).

Furthermore, fault-tolerant logic can only execute a restricted set of native operations (Clifford gates) easily. Non-Clifford operations, such as the arbitrary rotations required by QAOA layers, must be synthesized using specialized *T-gates*[17]. These gates cannot be executed directly and must be generated via a resource-intensive process known as *Magic State Distillation*[2]. The time required to distill these states (t_{distill}) dictates the clock speed of logical steps, transforming the flat runtime profile of FTQC into a function of the required mathematical accuracy.

3 The Max-Cut Problem and Algorithmic Solvers

To evaluate the operational ETS across classical and quantum architectures, a benchmark optimization task must be formally isolated. This framework utilizes the Max-Cut problem, a foundational NP-hard combinatorial optimization problem. This section defines the mathematical structure of Max-Cut and details the scaling behaviors of its dominant classical and quantum algorithmic solvers.

3.1 Mathematical Definition of Max-Cut

Let $G = (V, E)$ be an unweighted, undirected graph where $V = \{1, 2, \ldots, N\}$ represents the set of vertices (nodes) and E represents the set of edges connecting them. The goal of the Max-Cut problem is to partition the vertices into two disjoint subsets, S and $\bar{S}$ (where $\bar{S} = V \setminus S$), such that the number of edges crossing the boundary between the two sets is maximized.

By assigning a binary configuration variable $x_i \in \{-1, +1\}$ to each vertex $i \in V$ (indicating its membership in either S or $\bar{S}$), the problem can be framed as maximizing the following scalar cost function:

$$C(x) = \frac{1}{2} \sum_{(i,j) \in E} (1 - x_i x_j) \tag{3}$$

For a graph of size N, the solution space contains 2^N possible binary configurations, making a brute-force exhaustive search computationally intractable as N grows.

3.2 Classical Solvers and Computational Complexity

The computational footprint of solving Max-Cut classically depends strictly on the target approximation quality (measured by the approximation ratio $AR = Cut_{found}/Cut_{max}$, where Cut_{found} is the solution found, and Cut_{max} is the real MaxCut value), establishing a fundamental trade-off between energy expenditure and accuracy.

3.2.1 Exact Solvers: Exponential Scaling

To find the absolute global optimum (an Approximation Ratio $AR = 1.0$), classical frameworks must deploy exact global optimization techniques such as Branch-and-Bound or Branch-and-Cut. Because Max-Cut belongs to the NP-hard complexity class, the worst-case execution time for any exact classical algorithm scales exponentially with the number of variables:

$$t_{\text{exact}}(N) = \alpha_{\text{exact}} \cdot 2^N \tag{4}$$

where α_{exact} is a hardware-dependent constant. This exponential time complexity creates an insurmountable power wall for classical hardware at relatively small problem sizes (e.g., $N > 100$), where the total energy consumption yields an infinite vertical trajectory.

3.2.2 The Goemans-Williamson Algorithm: Polynomial Scaling

To bypass the exponential time wall, modern classical architectures rely on polynomial-time approximation algorithms. The definitive mathematical baseline is the Goemans-Williamson (GW) algorithm [8], which relaxes the discrete variables $x_i \in \{-1, +1\}$ into continuous, higher-dimensional unit vectors $v_i \in \mathbb{R}^N$. This converts the intractable discrete optimization problem into a continuous Semidefinite Program (SDP):

$$\text{Maximize } \frac{1}{2} \sum_{(i,j)\in E} (1 - v_i \cdot v_j) \quad \text{subject to } \|v_i\|^2 = 1. \tag{5}$$

Once solved, the high-dimensional vectors are mapped back to discrete binary cuts using a random hyper-plane slicing technique. The GW algorithm provides a strict mathematical guarantee: for any graph, it will achieve an expected approximation ratio of at least $AR_{\text{GW}} \approx 0.87856$[8].

3.3 The Quantum Approach: Quantum Approximate Optimization Algorithm

The Quantum Approximate Optimization Algorithm (QAOA) [22] is a hybrid quantum-classical variational algorithm designed to approximate the ground state of a problem Hamiltonian. For Max-Cut [20, 4, 6, 10, 5], the classical cost function is mapped directly onto a diagonal Ising Hamiltonian H_C by mapping the classical binary variables to quantum Pauli-Z operators ($x_i \to \sigma_i^z$):

$$H_C = \frac{1}{2} \sum_{(i,j)\in E} (I - \sigma_i^z \sigma_j^z) \tag{6}$$

The algorithm initializes an N-qubit system into a uniform superposition and alternately applies two non-commuting unitary operators over p sequential layers: the problem mixer $U(H_C, \gamma_k) = e^{-i\gamma_k H_C}$ and the transverse-field driver mixer $U(H_B, \beta_k) = e^{-i\beta_k H_B}$, where $H_B = \sum_{i=1}^N \sigma_i^x$. The resulting parameterized state is expressed as:

$$|\gamma, \beta\rangle = \prod_{k=1}^{p} e^{-i\beta_k H_B} e^{-i\gamma_k H_C} |+\rangle^{\otimes N} \tag{7}$$

The quantum processor executes this circuit layout, measures the state to extract sample shots (M_{shots}), and passes the expected energy ($\langle\gamma, \beta|H_C|\gamma, \beta\rangle$, which is equal to the expected cut value as given by Eq. 6) value back to a classical outer-loop optimizer, applying for instance a gradient ascent, moving the parameter along the direction of the steepest gradient. The classical loop iteratively tunes the $2p$ variational angles (γ, β) to maximize the cut quality.

For a fixed target accuracy, the execution runtime of QAOA is completely dependent on the layer depth p. To match the Goemans-Williamson approximation ratio baseline ($AR \approx 0.878$), analytic bounds on standard sparse graphs dictate that QAOA requires an operational depth of $p \geq 11$ [21]. Following the conjecture that under fixed optimal angles ([21], the AR will be larger than the performance guarantee (which for $p = 11$ is almost AR_{GW} for 3 regular graphs), we don't consider the step of finding the optimal angles. In the NISQ era, this depth incurs an exponential sampling (Eq. 1) overhead due to error mitigation, whereas in the FTQC era, it incurs a linear runtime increase governed by the continuous distillation of the non-Clifford T-gates required for applying the $2p$ rotations of QAOA.

4 Methodology

To establish a mathematically rigorous comparison between classical and quantum computing paradigms, this framework shifts the benchmarking metric away from raw execution time toward total energetic footprint. This section formalizes the core Energy-to-Solution (ETS) formulation, details the systemic boundary conditions used to isolate power consumption, and establishes the accuracy-parity constraint that governs the scaling plots presented in Section 5. We follow mostly the methodology developed and used in [7], as they also analyze both NISQ and FTQC case, coming up with the equations needed for estimating the power consumption of each, in order to study the same question of whether quantum computing is energetically efficient or not. In this sense, by incorporating some of their equations (Appendix B), we also adopt the same assumptions regarding the hardware and its estimations adopted by them.

4.1 The Energy-to-Solution (ETS) Formulation

In classical data centers and quantum computing facilities alike, transient power draw fluctuate dynamically based on computational load. However, the true environmental and operational cost of solving a specific problem instance is determined by the total integrated energy consumed over the duration of the calculation.

For a given graph size N and a fixed target solution accuracy (AR), the ETS is modeled as the direct product of the effective systemic power consumption (P) and the total computational execution time (t):

$$ETS(N) = P_{\text{systemic}} \cdot t_{\text{execution}}(N). \tag{8}$$

where ETS is measured in Joules (J), P_{systemic} represents the continuous operational wattage (W), and $t_{\text{execution}}$ represents the runtime in seconds (s) required for the specific architecture to terminate and output the solution.

4.2 Power Quantification Parameters by Architecture

The power variable P_{systemic} is uniquely formalized for each of the three evaluated cases to reflect the physical reality of their respective infrastructure stacks:

1. **Classical GPU Case:** The power footprint models an enterprise-grade accelerated computing node (we analyze the case of a NVIDIA H100). The systemic power is calculated as the sum of the dedicated GPU Thermal Design Power and the supporting host CPU overhead, scaled directly by the facility's Power Usage Effectiveness (PUE):

$$P_{\text{classical}} = (P_{\text{GPU}} + P_{\text{host}}) \cdot \text{PUE} \tag{9}$$

2. **NISQ Case:** The power model captures a localized superconducting quantum processing unit (QPU). The baseline draw is dominated by the static, continuous cryogenic infrastructure (P_{fridge}) required to maintain the dilution refrigerator at millikelvin temperatures, supplemented by the dynamic, room-temperature control electronics (P_{control}) scaling linearly with the physical qubit count:

$$P_{\text{NISQ}} = P_{\text{fridge}} + \Big(N \cdot P_{\text{control/qubit}}\Big) \tag{10}$$

   In our calculations, we neglect the second term because P_{fridge} dominates the total power.[7]

3. **FTQC Case:** The fault-tolerant power profile accounts for massive scaling. It integrates an expanded macroscopic cryogenic infrastructure ($P_{\text{macro\_fridge}}$) alongside an aggressive classical computing allocation dedicated to real-time topological error correction. This is governed by the syndrome decoding power (P_{decoder}), which processes data from many physical qubits:

$$P_{\text{FTQC}} = P_{\text{macro\_fridge}} + \Big(N_{\text{phys}} \cdot P_{\text{decoder/qubit}}\Big) \tag{11}$$

4.3 The Accuracy-Parity Constraint and Algorithmic Runtime

A critical structural vulnerability in comparative benchmarking is the comparison of unequal algorithmic outputs. A fast heuristic achieving a poor-quality cut cannot be directly compared to a slow, near-exact optimization. To ensure strict fairness, all analytical scaling plots generated in this framework are constrained to evaluate the execution time $t(N)$ required to reach an identical, fixed target approximation ratio AR.

To build the scaling plots, our script takes the problem size N (number of graph vertices) as the independent variable on the X-axis and compute the runtime $t(N)$ for each case. The runtime functions are derived directly from the abstract number of mathematical or physical operations required by each solver to meet this target accuracy.

By mapping these disparate algorithmic operational counts, the resulting crossover curves reveal the exact physical threshold where quantum architectures achieve a true, systemic energy breakeven point over state-of-the-art classical supercomputing nodes.

Here are the specific runtime scaling functions for Max-Cut solved across three paradigms:

4.3.1 Classical Heuristic Runtime Function

To keep the comparison realistic, we don't focus on a brute-force exact solver (which scales as $O(2^N)$ and would lose to a quantum computer almost instantly) and instead model a high-performance classical heuristic approximation algorithm, the Goemans-Williamson algorithm[8].

To translate the number of algorithmic operations into physical execution seconds, we simply divide the total algorithmic workload by the processing speed of the hardware. The standard metric for hardware speed is FLOPS (Floating-Point Operations Per Second). Because the Goemans-Williamson algorithm relies on Semidefinite Programming (SDP), which requires high mathematical precision to converge properly, we must use the Double Precision FLOPS (FP64) specification of the hardware. Here is the formula for the classical runtime baseline:

$$t_{\text{classical}}(N) = \frac{C \cdot N^{3.5}}{\text{FLOPS}_{\text{peak}} \cdot E_{\text{util}}} \tag{12}$$

- C: The algorithm's constant multiplier. For matrix-heavy SDP solvers, the overhead multiplier is typically around 100;
- $\text{FLOPS}_{\text{peak}}$: For an H100, the peak FP64 performance using Tensor Cores is roughly 67 TFLOPS (67×10^{12} operations per second)[15];
- E_{util}: Hardware utilization efficiency. GPUs almost never hit their peak theoretical speed because they waste time moving data in and out of memory. For complex SDP solvers, a realistic utilization efficiency is 10% to 20% (0.1 to 0.2).

4.3.2 NISQ Runtime Function

In the NISQ era, physical qubits must execute gates directly. The physical circuit depth (D_{phys}) depends on how the graph layout maps to the constrained 2D grid of a superconducting chip.

Step A: Compute Physical Circuit Depth

For a Sparse Graph (e.g., 3-Regular): Qubits only need to interact with 3 neighbors. With smart scheduling, the depth scales independently of system size: $D_{\text{phys}}(N) \approx c_1 \cdot p$.[5]

For a Dense/Fully Connected Graph: Every qubit must interact with every other qubit, requiring a network of SWAP gates to move states across the chip. The physical depth scales linearly: $D_{\text{phys}}(N) \approx c_2 \cdot p \cdot N$.

In this work we only consider the first case, in which depth doesn't scale with problem size N, because it's where performance is better (since NISQ computations get highly noisy for larger depths, requiring error mitigation), and also because we are considering the fixed angles conjecture of [21] in our model for QAOA implementation, as commented in Section 3.3.

Step B: Apply the Quantum Error Mitigation (QEM) Penalty

To mitigate errors without active correction, techniques like Probabilistic Error Cancellation require an exponential explosion in the number of measurement samples (shots) as the circuit gets deeper. The runtime is then given by:

$$t_{\text{NISQ}}(N) = M_{\text{shots}} \cdot (D_{\text{phys}}(N) \cdot t_{\text{gate}}) \cdot e^{2 \cdot D_{\text{phys}}(N) \cdot \epsilon_{\text{phys}}} \tag{13}$$

- M_{shots}: The baseline number of shots needed to resolve the QAOA optimization landscape (typically fixed at 10^4);
- t_{gate}: The physical gate execution speed (following reference [7], we consider it to be the time of a 2 qubit gate, which is 4 times thar of a 1 qubit gate: ~ 100 ns);
- ϵ_{phys}: The physical gate error rate ($\sim 10^{-3}$);
- $e^{2D\epsilon}$: This is the core of the NISQ baseline and comes from Eq. 1. As N grows, this exponential scaling factor will cause the NISQ energy line to shoot up vertically.

4.3.3 FTQC Runtime Function

In a fault-tolerant system, physical errors are entirely suppressed by a surface code. However, QAOA requires arbitrary rotation angles (non-Clifford gates). In fault-tolerance, these rotations cannot be done natively; they must be synthesized using T-gates, which are produced slowly by "Magic State Distillation" factories.

Step A: Compute Logical Step Time

The time required to execute an arbitrary QAOA rotation step depends on the number of T-gates (N_T) needed to reach a target mathematical precision (δ).[7]

$$t_{\text{logical\_step}} = N_T \cdot t_{\text{distill}} \tag{14}$$

- $N_T \approx 3\log_2(1/\delta)$ (Typically ~ 30 to 50 T-gates per angle rotation);
- t_{distill}: The time required to distill a clean magic state through the surface code cycles ($\sim$ 10 to 20 μs).

Step B: Total Fault-Tolerant Runtime

Because errors are actively corrected, there is no exponential sampling penalty. The system only needs a static, baseline number of optimization shots (M_{FTQC}) to converge.

$$t_{\text{FTQC}}(N) = M_{\text{FTQC}} \cdot p \cdot L_{\text{depth}}(N) \cdot t_{\text{logical\_step}} \tag{15}$$

- $L_{\text{depth}}(N)$: The logical circuit depth. For sparse graphs, it is a constant $O(1)$ relative to N due to massive parallel execution of logical gates. For fully connected graphs, it scales as $O(N)$, but we don't analyze it;
- M_{FTQC}: Fixed baseline shots (e.g., 10^3).

5 Results

The result of our analysis are summarized into two graphs plotted in Figure 1 and 2. Both are related to the same scenario, in which we fixed our final accuracy for all the methods studied, and compared the number of operations each of them need to obtain such accuracy. A fixed accuracy is necessary, as described in previous sections, in order to the comparison between algorithms to make sense. Notice that we keep the Brute-Force algorithm in the plots even though it has $AR = 1$, that is, it's an exact solution. This only to explicit its inefficiency with respect to the others.

In the first one, we have the plot of the energy cost for each solution, our Energy-to-Solution defined in Section 4.1, in function of the problem complexity, N, which corresponds to the size of the graph that we are trying to find the Max-Cut.

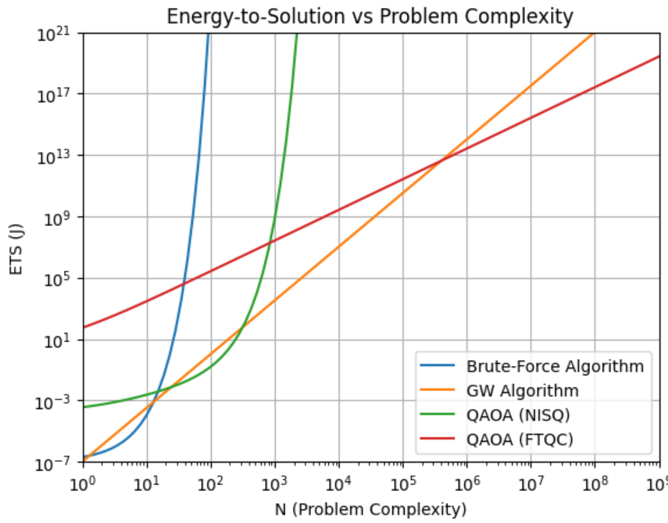

Figure 1: Plot of the Energy-to-Cost found for each case.

As expected, we see in Figure 1 that quantum hardware has a greater initial cost, while the classical counterpart is practically zero. But as the problem size grows, we see a change, and we can spot the break-even points that we were interested in, which are exactly the moments where QAOA algorithm (NISQ or FTQC) becomes less energetically expensive than the classical ones. For the NISQ case, it becomes more advantageous than the GW algorithm earlier than the FTQC (thanks to its lower initial cost). But, as a consequence of the necessity to sustain the target GW approximation ratio ($AR \approx 0.878$) which demands a deeper variational circuit, and therefore more errors occur, we observe an exponential grow related to Quantum Error Mitigation. For this reason, NISQ becomes less efficient than FTQC and GW at a later point.

So this indicates the fact that near-term quantum architectures cannot scale sustainably for high-complexity optimization problems, as errors get too big and mitigating them is too costly. However, for medium size problems, NISQ can be interesting, as these results show they are able to obtain the same result than classical methods consuming less energy. Unfortunately, in this work we don't analyze if quantum algorithms allows us to obtain better results than classical, since we fixed our accuracy to compare with the best accuracy that GW can get.

In the other hand, for the FTQC case, we see that we don't have anymore the exponential grow of NISQ devices, because topological surface codes suppress physical errors exponentially, completely bypassing the sampling tax of error mitigation, resulting in a remarkably linear runtime scaling, allowing us to achieve the best performance of the quantum algorithm. The negative side is that FTQC have a high initial energy cost driven by a continuous cryogenic infrastructure overhead and the heavy real-time power draw of classical syndrome decoders processing millions of physical qubits. So it starts with a very high initial cost, and becomes more efficient than GW only in a later moment. This is another break-even point of interest in this work. The interesting thing about this one is that after this point, FTQC will always be energetically better than the classical counterpart. Since it's for bigger problems that this energy difference becomes more relevant, this is a great advantage for FTQC, and is one example of why we have a global interest in the development of quantum computers.

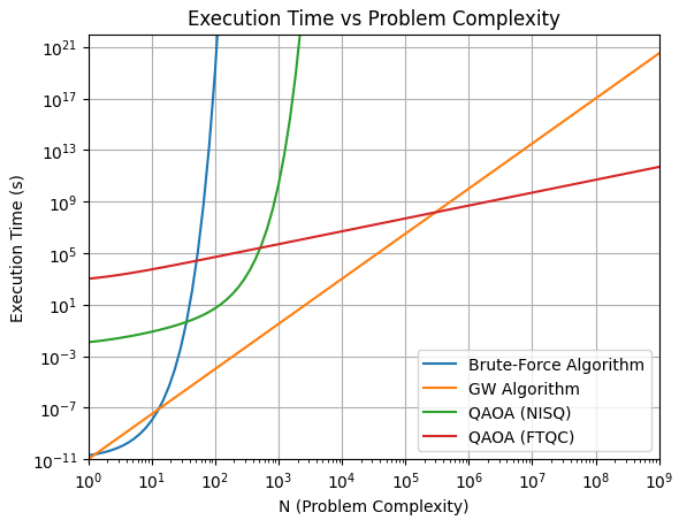

Figure 2: Plot of the execution time for each algorithm.

The second Figure (2) show the time of execution of each algorithm. It's directly related to the first one that we discussed above, as we can see that both the Brute-Force and the QAOA (NISQ) are exponential in Figure 1 because of the exponential growth in their time of execution. Similarly, GW and QAOA (FTQC) are linear because they have linear growth in execution time. Note that we are referring to exponential and linear growth with respect to the log scale of both Figures.

In Figure 2 we notice that QAOA (NISQ) is always slower than GW algorithm, even though we found that there is an interval of N in which NISQ is more efficient than GW. So it highlights something central throughout this project, which is the idea of quantum energy advantage, instead of the usual quantum advantage based on execution time of algorithms. In the usual sense of quantum advantage, QAOA in a NISQ device didn't show any advantage in our analysis compared to the best classical algorithm, even though it shows a window of quantum energy advantage in terms of problem complexity.

Going back to FTQC, we see that it also starts with a long initial execution time, but then we see it becomes faster than GW, indicating here a quantum advantage in the sense of speed-up in the computation. We see though that it becomes energetically better than GW a bit earlier in terms of problem complexity. Therefore, we notice that quantum computers may present energy advantages before presenting any computational advantage, or even without ever presenting it, as for the case of NISQ seen above.

6 Conclusion

This research project has successfully established and evaluated a comprehensive analytical Break-even Analysis framework to investigate the conditions under which quantum computing can deliver a systemic *Quantum Energy Advantage* for a combinatorial optimization (Max-Cut). By prioritizing the holistic metric of Energy-to-Solution (ETS) at a strict accuracy-parity threshold (i.e. a fixed approximation ratio), this work bypasses the traditional benchmarking trap of comparing different computing architectures without holding solution accuracy and problem parameters constant, providing a mathematically rigorous foundation for comparative sustainable computing. The analytical scaling plots generated for the Max-Cut problem reveal three highly distinct architectural trajectories as problem size (N) scales: the Classical GPU Baseline (Goemans-Williamson) operating on state-of-the-art enterprise hardware (NVIDIA H100 system), the NISQ scaling wall, and the FTQC paradigm.

Our modeling reveals that the flat FTQC curve successfully intersects the rising polynomial Classical curve at a definitive breakeven threshold of $N \approx 5 \cdot 10^5$ nodes, representing a total energy cost of $ETS \approx 10^{12}$ Joules. Beyond this critical computational boundary, the high infrastructure overhead of fault-tolerant systems is entirely offset by its superior scaling efficiency, rendering FTQC the only viable path toward an environmentally sustainable "green" computational advantage.

Even though these results are accordingly to what we expected them to be, they are not completely precise, so we cannot assure that the break-even points found correspond to good estimations of the real ones that we would find in practice. The models and assumptions for each case simplify a lot of details that might play a more significant role than expected, changing these results. This project puts together many state-of-the-art topics of research, and some of the information needed to obtain these results are yet not so well established, such as QAOA implementation for Max-Cut and its complexity (number of gates, circuit depth), FTQC, magic state distillation, and energy cost of error-correcting, superconducting quantum computers parameters and energetic costs. Thus, this work should be seen as a first estimation of the actual energy costs of the algorithms analyzed.

Therefore, while the analytical framework developed in this project provides a clear architectural roadmap, several simplifying assumptions present fertile ground for future research. First, this model assumed a uniform classical syndrome decoding power overhead; in a physical system, the decoding workload scales with the code distance d, which must increase to maintain logical integrity for exceptionally deep circuits. Second, our quantum model focused exclusively on superconducting transmons. Future iterations of this framework should integrate alternative physical modalities, such as neutral-atom or photonic systems, which operate at higher ambient temperatures and may fundamentally lower the initial cryogenic entry fee, shifting the energy breakeven threshold significantly to the left. Ultimately, as global computing demands continue to strain the limits of silicon-based infrastructure, the refinement of these physical-algorithmic energy models will remain vital to guiding the sustainable scaling of the next generation of high-performance computers.

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A Qubits, Superposition, and Mathematical Representation

[1]Unlike a classical bit, which is restricted to a discrete state $x \in \{0, 1\}$, the fundamental unit of quantum information is the quantum bit, or *qubit*. A physical qubit is a two-level quantum system described mathematically as a vector in a two-dimensional complex Hilbert space $\mathbb{H}^2$. The state of a single qubit $|\psi\rangle$ can exist in a linear combination of the computational basis states $\{|0\rangle, |1\rangle\}$, expressed as:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \tag{16}$$

where $\alpha, \beta \in \mathbb{C}$ represent probability amplitudes constrained by the normalization condition $|\alpha|^2 + |\beta|^2 = 1$. When the qubit is measured in the computational basis, the superposition collapses, yielding the classical outcome 0 with probability $|\alpha|^2$ and 1 with probability $|\beta|^2$.

[1]This Appendix should be more completed, introducing evolution of states, measurements and some ideas.

For an N-qubit system, the state space scales exponentially as a tensor product $\mathcal{H} = (\mathbb{H}^2)^{\otimes N}$, spanning a 2^N-dimensional Hilbert space. It is this exponential state space that allows quantum algorithms to process highly complex optimization landscapes, such as the Max-Cut problem, without storing an explicit 2^N-element vector in classical memory.

B Power estimation

[2]The power P_π consumed to bring the qubit from the ground state $|0\rangle$ to the excited state $|1\rangle$ of the superconducting qubit is the given by:

$$P_\pi = \frac{\hbar\omega_0\pi^2}{4\gamma\tau_{1qb}^2} \tag{17}$$

Where ω_0 is the transition frequency from the ground state $|0\rangle$ to the excited state $|1\rangle$ ($\omega_0 \approx 2\pi \times 6$ GHz), τ_{1qb} is the pulse duration, γ is the spontaneous emission rate ($\gamma \approx 1000s$).

For model simplicity, all single-qubit gates are taken as X gates, so that P_pi is the power consumed by any single qubit gate.

The cryogenic electrical power consumption needed to actually run a gate (considering that the qubit is in a cryogenic temperature) is given by:

$$P_{1qb} = \frac{T_{ext} - T_{qb}}{T_{qb}} AP_\pi \tag{18}$$

Where T_{ext} is the external temperature ($\approx 300\,K$), T_{qb} is the temperature of the qubits ($\approx 0.1\,K$), and A is an attenuation factor.

[2]This Appendix is also not completed, although it contains two important equations for estimating the power of a quantum gate.